Four weekly sprints: build, test and ship every day

SPRINT 1 · Day 01-07 · 27 Oct - 2 Nov

Foundations & secure access

A secured AWS base and an authenticated API that talks to Bedrock and records every request.

DELIVERABLES

- Repo, remote state, budget alerts, CI
- Identity Center + sandbox target account
- Authenticated POST /prompt calling Bedrock (Claude)
- DynamoDB request and audit lifecycle

EXIT CRITERIA

- Token in, Bedrock reply out, request stored
- CI green; spend alerts confirmed

SPRINT 2 · Day 08-14 · 3 Nov - 9 Nov

Agent brain

Turn a plain-English prompt into validated Terraform for UC1-UC3, safely.

DELIVERABLES

- System prompt + structured plan schema
- Tool-use loop: generate, validate, summarise
- Knowledge base of approved modules (RAG)
- Bedrock Guardrails + self-correction loop

EXIT CRITERIA

- UC1-UC3 prompts produce Terraform that passes validate (target: 90% of a 20-prompt eval set)
- Out-of-scope prompts blocked

SPRINT 3 · Day 15-21 · 10 Nov - 16 Nov

Delivery pipeline

An approved prompt becomes deployed infrastructure, with security, cost and human gates.

DELIVERABLES

- Step Functions workflow with failure paths
- CodeBuild plan + policy gate + cost gate
- Slack/SNS approval with task token
- Cross-account apply to the sandbox

EXIT CRITERIA

- UC1-UC3 deploy end to end through approval
- N1, N2 and N3 behave as designed; teardown works

SPRINT 4 · Day 22-30 · 17 Nov - 25 Nov

Hardening & launch

Deploy the headline 2-tier app from one prompt, make it resilient and auditable, then ship it.

DELIVERABLES

- ALB, ECS Fargate and RDS modules; UC4 + UC5
- Rollback, TTL auto-destroy, drift detection
- Dashboards, audit view, security review
- Chat UI, demo video, final write-up

EXIT CRITERIA

- UC5 deploys from a single prompt
- All UC and N tests pass; fresh clone follows README

USE-CASE TEST LADDER UC = what the agent must build | N = what it must refuse or stop

UC1 · Encrypted S3 bucket Generate D10 | Deploy D20	**UC2 · VPC with public/private subnets** Generate D10 | Deploy D21	**UC3 · Lambda + API Gateway + DynamoDB** Generate D13 | Deploy D21	**UC4 · ECS Fargate behind an ALB** Plan + deploy D22	**UC5 · 2-tier app: ALB + ECS + RDS** Headline demo D23
N1 · Public S3 request is blocked Policy gate D17	**N2 · Cost over limit is flagged** Cost gate D18	**N3 · Human rejects: nothing deployed** Approval D19	**N4 · Injection / out-of-scope prompt blocked** Guardrails D12	

Every day: Build → Test → Ship (commit, CI green, LinkedIn post)

Day 00 was the kickoff on 26 Oct

SPRINT 1

Foundations & secure access

Day 01-07 · 27 Oct - 2 Nov

Goal: A secured AWS base and an authenticated API that talks to Bedrock and records every request.

DAY 01 — Tue 27 Oct

Repo, remote state, budget

BUILD
Terraform bootstrap (KMS-encrypted S3 state), AWS Budget, CI with fmt, validate and Checkov.

TEST
terraform plan and apply clean; CI green on a pull request; budget email confirmed.

SHIP
Public repo + Day 01 post.

DAY 02 — Wed 28 Oct

Accounts & identity

BUILD
IAM Identity Center; second AWS account as sandbox target; permission boundary.

TEST
SSO login works in both accounts; creating resources outside the boundary is denied.

SHIP
Account layout diagram.

DAY 03 — Thu 29 Oct

Bedrock hello-agent

BUILD
Enable Claude model access; Lambda calls Bedrock Converse API; Terraform module.

TEST
Unit test with mocked client; live invoke returns text.

SHIP
First model call demo.

DAY 04 — Fri 30 Oct

API + authentication

BUILD
API Gateway + Cognito user pool; POST /prompt invokes the Lambda.

TEST
401 without token, 200 with a valid JWT; 20 quick requests succeed.

SHIP
Authenticated endpoint.

DAY 05 — Sat 31 Oct

Request data model

BUILD
DynamoDB requests and audit table, KMS, TTL; status lifecycle RECEIVED to DEPLOYED.

TEST
Read/write unit tests; an illegal status change is rejected.

SHIP
Lifecycle diagram.

DAY 06 — Sun 1 Nov

Logging & alarms

BUILD
Structured logs with correlation ID; CloudWatch dashboard and error alarm.

TEST
Forced error trips the alarm; one ID traces a request end to end.

SHIP
Dashboard screenshot.

DAY 07 — Mon 2 Nov

Sprint 1 review

BUILD
ADRs, cleanup, cost check, smoke-test script.

TEST
Smoke test: token to /prompt to DynamoDB record; CI green.

SHIP
Week 1 recap + demo video.

Sprint 1 review

DELIVERABLES

- Repo, remote state, budget alerts, CI
- Identity Center + sandbox target account
- Authenticated POST /prompt calling Bedrock (Claude)
- DynamoDB request and audit lifecycle

EXIT CRITERIA

- Token in, Bedrock reply out, request stored
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Targets are goals, not results; update them as the build teaches us more

SPRINT 2

Agent brain

Day 08-14 · 3 Nov - 9 Nov

Goal: Turn a plain-English prompt into validated Terraform for UC1-UC3, safely.

DAY 08 — Tue 3 Nov

Prompt design

BUILD
System prompt and JSON plan schema (resources, region, tags).

TEST
20-prompt eval set; every plan passes schema validation.

SHIP
Prompt v1 + eval set.

DAY 09 — Wed 4 Nov

Tool use

BUILD
Converse tools: generate_terraform, validate_terraform, summarise_plan.

TEST
Mocked tool-loop tests; tool arguments schema-checked.

SHIP
Tool schema doc.

DAY 10 — Thu 5 Nov

Generate UC1 + UC2

BUILD
Terraform for an encrypted S3 bucket and a VPC with subnets.

TEST
terraform fmt and validate pass on both outputs.

SHIP
UC1 + UC2 demo.

DAY 11 — Fri 6 Nov

Knowledge base (RAG)

BUILD
Approved modules, naming and tagging standards in a Bedrock knowledge base.

TEST
Approved module in the top 3 results for 10 queries; code uses the module.

SHIP
KB design post. Cost guard: tear down or pick a cheaper vector store.

DAY 12 — Sat 7 Nov

Guardrails (N4)

BUILD
Bedrock Guardrails: block out-of-scope and injection prompts.

TEST
10 malicious prompts blocked, 10 valid prompts pass.

SHIP
Guardrail results.

DAY 13 — Sun 8 Nov

Self-correction + UC3

BUILD
Feed validate errors back to the model, max 3 retries; generate Lambda + API + DynamoDB.

TEST
Injected errors fixed within 3 tries; retry cap enforced; UC3 validates.

SHIP
Before/after of the fix loop.

DAY 14 — Mon 9 Nov

Sprint 2 review

BUILD
Eval report, prompt v2, tidy-up.

TEST
Re-run the 20-prompt eval and record the pass rate; add regression test to CI.

SHIP
Week 2 recap + metrics.

Sprint 2 review

DELIVERABLES

- System prompt + structured plan schema
- Tool-use loop: generate, validate, summarise
- Knowledge base of approved modules (RAG)
- Bedrock Guardrails + self-correction loop

EXIT CRITERIA

- UC1-UC3 prompts produce Terraform that passes validate (target: 90% of a 20-prompt eval set)
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SPRINT 3

Delivery pipeline

Day 15-21 · 10 Nov - 16 Nov

Goal: An approved prompt becomes deployed infrastructure, with security, cost and human gates.

DAY 15 — Tue 10 Nov

Step Functions workflow

BUILD
State machine: generate, validate, gate, approve, apply, notify, with catch states.

TEST
Mocked-task execution; a failure ends in FAILED with a reason.

SHIP
State machine diagram.

DAY 16 — Wed 11 Nov

Plan stage

BUILD
CodeBuild runs terraform init and plan; plan JSON saved to S3.

TEST
UC1 plan produced and stored; no apply happens.

SHIP
Plan artifact.

DAY 17 — Thu 12 Nov

Policy gate (N1)

BUILD
Checkov + OPA: encryption, no public access, required tags, allowed regions.

TEST
Public-S3 request blocked; compliant request passes; 15 policy tests.

SHIP
Policy-as-code folder.

DAY 18 — Fri 13 Nov

Cost gate (N2)

BUILD
Infracost on the plan JSON; monthly threshold in config.

TEST
Over-limit request flagged; UC2 estimate matches the Infracost CLI.

SHIP
Cost summary in the approval message.

DAY 19 — Sat 14 Nov

Human approval (N3)

BUILD
Slack/SNS summary (resources, cost, risks); approve/reject via task token.

TEST
Reject deploys nothing; approve continues; timeout expires the request.

SHIP
Approval flow demo.

DAY 20 — Sun 15 Nov

Cross-account apply

BUILD
Deployer role in target account (ExternalId, least privilege); CodeBuild apply; outputs returned.

TEST
UC1 deployed in sandbox; role cannot act outside allowed services; state saved.

SHIP
First end-to-end deploy.

DAY 21 — Mon 16 Nov

Sprint 3 review

BUILD
Run UC1-UC3 end to end; fix gaps.

TEST
Regression: UC1-UC3 + N1-N3 pass; terraform destroy cleans up.

SHIP
Week 3 recap + demo video.

Sprint 3 review

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- Step Functions workflow with failure paths
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SPRINT 4

Hardening & launch

Day 22-30 · 17 Nov - 25 Nov

Goal: Deploy the headline 2-tier app from one prompt, make it resilient and auditable, then ship it.

DAY 22 — Tue 17 Nov

Complex templates (UC4)

BUILD
ALB, ECS Fargate and RDS modules in the KB; multi-resource planning.

TEST
UC4 plan validates and passes the policy gate; dependency order correct.

SHIP
UC4 plan.

DAY 23 — Wed 18 Nov

Headline deploy (UC5)

BUILD
Wire the 2-tier app: ALB + ECS Fargate + RDS Postgres.

TEST
Deploy to sandbox; ALB URL returns 200; RDS private and encrypted.

SHIP
UC5 working demo.

DAY 24 — Thu 19 Nov

Failure handling & rollback

BUILD
Retries, idempotency, automatic rollback on partial failure.

TEST
Inject a mid-apply failure: state consistent, user notified, no orphan resources.

SHIP
Failure demo.

DAY 25 — Fri 20 Nov

Auto-destroy & drift

BUILD
TTL destroy via EventBridge Scheduler; drift check with plan exit code.

TEST
10-minute TTL stack is destroyed; injected drift is detected.

SHIP
Cost-saving post.

DAY 26 — Sat 21 Nov

Observability & audit

BUILD
Dashboards (success rate, latency, cost per request); CloudTrail audit view.

TEST
One query returns prompt, code, plan, approver, result; alarm fires on failure.

SHIP
Dashboard screenshot.

DAY 27 — Sun 22 Nov

Security review

BUILD
Tighten IAM with Access Analyzer; secrets scan; threat-model notes.

TEST
Access Analyzer findings fixed or documented; no secrets found; Checkov green.

SHIP
Security write-up.

DAY 28 — Mon 23 Nov

Simple chat UI

BUILD
Minimal web UI with Cognito login calling /prompt.

TEST
Browser run of UC1 and UC5 end to end; phone-width check.

SHIP
UI demo.

DAY 29 — Tue 24 Nov

Full regression + demo

BUILD
Run UC1-UC5 and N1-N4 together; record the demo.

TEST
Every test passes; total AWS cost for the project recorded.

SHIP
Demo video.

DAY 30 — Wed 25 Nov

Launch

BUILD
Final README, architecture update, lessons learned, cost report; teardown.

TEST
A fresh clone sets up by following the README alone.

SHIP
Launch post + repo.

Sprint 4 review

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